

Math Sec 1

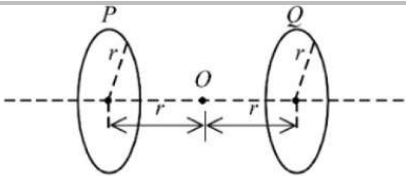
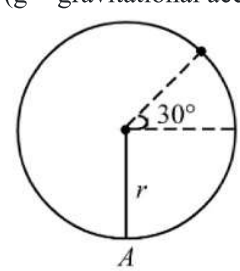
1. The smallest positive integral value of a , for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to
(A) 9 (B) 7 (C) 3 (D) 4
2. Consider the following three statements for the function $f : (0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = |\log_e x| - |x - 1|$:
(I) f is differentiable at all $x > 0$.
(II) f is increasing in $(0, 1)$.
(III) f is decreasing in $(1, \infty)$.
Then.
(A) Only (I) and (III) are TRUE.
(B) Only (II) and (III) are TRUE.
(C) Only (I) is TRUE.
(D) All (I), (II) and (III) are TRUE.
3. Let f be a function such that $3f(x) + 2f(\frac{m}{19x}) = 5x$, $x \neq 0$, where $m = \sum_{i=1}^9 i^2$. Then $f(5) - f(2)$ is equal to
(A) 9 (B) 18 (C) 36 (D) -9
4. Q. $4 \left(\frac{1}{3} + \frac{4}{7} \right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2} \right) + \left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3} \right) + \dots$ upto infinite terms, is equal to
(A) $\frac{4}{3}$ (B) $\frac{7}{4}$
(C) $\frac{6}{5}$ (D) $\frac{5}{2}$
5. Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj } P))$ is:
(A) 32 (B) 81 (C) 124 (D) 16
6. The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is
(A) 1580 (B) 1581 (C) 1579 (D) 1578
7. Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}) \cdot \vec{c} = -97$, then $|\vec{c} \times \hat{k}|^2$ is equal to
(A) 233 (B) 205 (C) 193 (D) 218
8. Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to
(A) 23 (B) 21 (C) 24 (D) 27
9. Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in \mathbb{N}$. Then $a + b + c$ is equal to
(A) 4 (B) 8 (C) 12 (D) 6
10. Let $X = \{x \in \mathbb{N} : 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{F}$ define $Y = \{ax + b : x \in X\}$. If the mean and variance of the elements of Y are 30 and 750 respectively, find the sum of all possible values of b .
(A) 20 (B) 100 (C) 80 (D) 60
11. Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$), be 30.
If its eccentricity is the maximum value of the function $f(t) = -\frac{3}{4} + 2t - t^2$, then $(a^2 + b^2)$ is equal to
(A) 516 (B) 276 (C) 256 (D) 496
12. Let $[t]$ denote the greatest integer less than or equal to t . If the function $f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2} \left[\frac{x}{2} (\cos x + \sin x) \cos x \right] \right), & x < 0 \\ \frac{\sin x - \frac{1}{2} \sin 2x}{x^3}, & x > 0 \end{cases}$ is continuous at $x = 0$, then $a^2 + b^2$ is equal to
(A) $\frac{1}{2}$ (B) $\frac{9}{16}$
(C) $\frac{3}{4}$ (D) $\frac{5}{8}$
13. Let $f(x) = \int \frac{7x^{10} + 9x^8}{(1+x^2+2x^9)^2} dx$, $x > 0$,
 $\lim_{x \rightarrow 0} f(x) = 0$ and $f(1) = \frac{1}{4}$. If $A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$ be such that $|B| = 81$, then α^2 is equal to
(A) 2 (B) 1 (C) 4 (D) 3
14. Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to:
(A) $\frac{\sqrt{35}}{2}$ (B) 7
(C) $\frac{\sqrt{21}}{2}$ (D) 13
15. Let $f(a)$ denote the area of the region in the first quadrant bounded by $x = 0$, $x = 1$, $y^2 = x$ and $y = |ax - 5| - |1 - ax| + ax^2$. Then $(f(0) + f(1))$ is equal to
(A) 12 (B) 7 (C) 9 (D) 14

16. Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim_{t \rightarrow x} \left(\frac{t^2 y(x) - x^2 y(t)}{x-t} \right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to
 (A) 18 (B) 12 (C) 23 (D) 27
17. If the domain of the function $f(x) = \sin^{-1} \left(\frac{1}{x^2 - 2x - 2} \right)$ is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to
 (A) 2 (B) 3 (C) 4 (D) 5
18. The largest value of n , for which 40^n divides $60!$, is
 (A) 13 (B) 14 (C) 11 (D) 12
19. The sum of all values of α , for which the shortest distance between the lines $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is
 (A) -8 (B) 6 (C) -6 (D) 8
20. Let the angles made with the positive x -axis by two straight lines drawn from the point $P(2, 3)$ and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2 . Then the value of $(\theta_1 + \theta_2)$ is:
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
 (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{2}$

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21. Let (h, k) lie on the circle $C : x^2 + y^2 = 4$ and the point $(2h+1, 3k+2)$ lie on an ellipse with eccentricity e . Then the value of $\frac{5}{e^2}$ is equal to _____.
22. If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$, then $e + f(0)$ is equal to _____.
23. The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ)\}$ is _____.
24. Let $z = (1+i)(1+2i)(1+3i) \dots (1+ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to _____.
25. Let S be a set of 5 elements and $P(S)$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $P(S) \times P(S)$ such that $A \cap B = \emptyset$. The probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in \mathbb{N}$, the value of $p+q$ is equal to _____.

Phy Sec 1

26. 
 Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O . The net magnetic field at O is:
 (A) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards P
 (B) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q
 (C) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P
 (D) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q
27. A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30° . Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is _____.
 (A) $\frac{\sqrt{3}}{2}mg$ (B) $\frac{1}{2}mg$
 (C) $\sqrt{3}mg$ (D) mg
28. A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will
 (A) decrease by 4 times (B) increase by 8 times
 (C) decrease by 8 times (D) increase by 64 times
29. The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is _____.
 (A) 1 (B) 2 (C) 3 (D) 4
30. In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is
 (g = gravitational acceleration)

 (A) $\sqrt{5gr}$ (B) $\sqrt{4gr}$